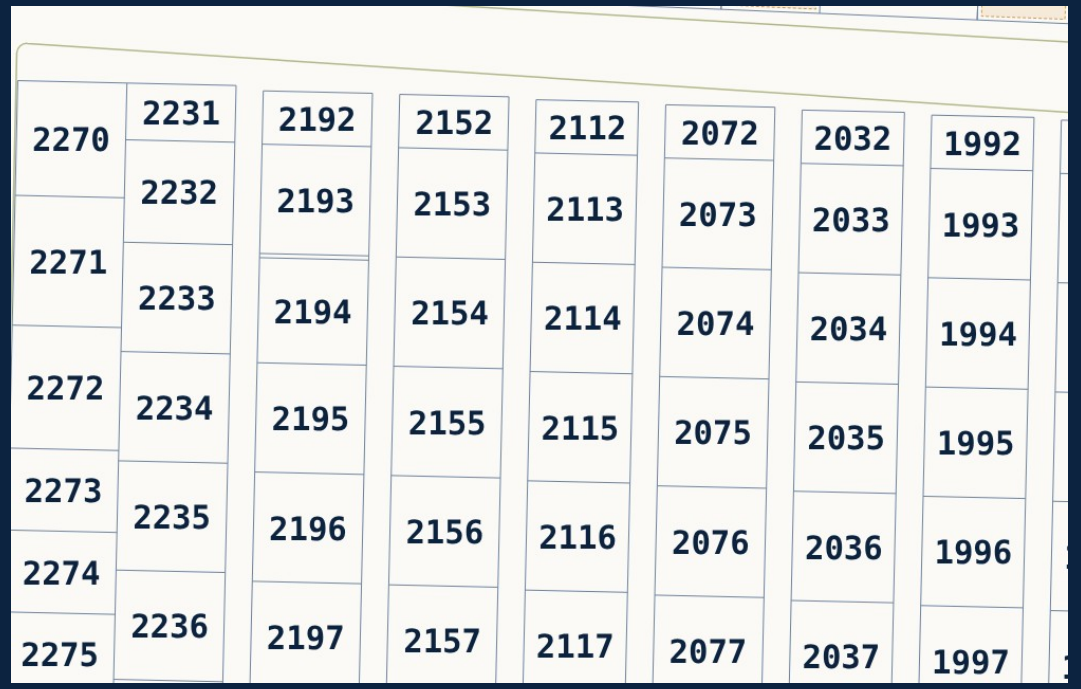
Cedar Grove H-Number Atlas

Map cleanup, H-number placement, and remaining records work

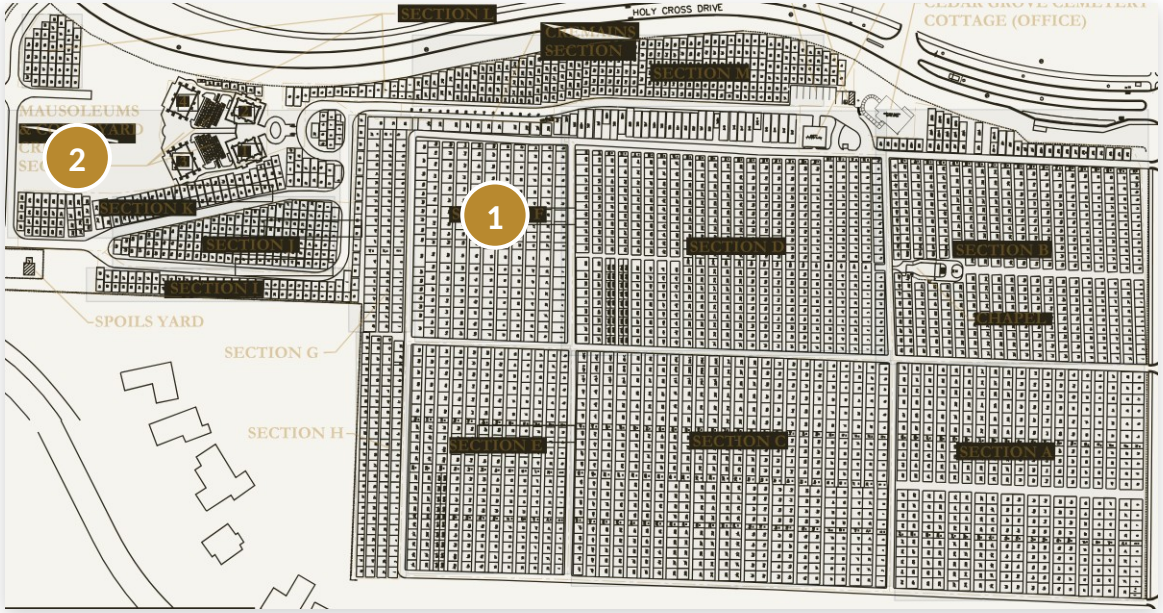


Current map: H-numbers inside their lot outlines

Mahmoud Hafez | October 12, 2026

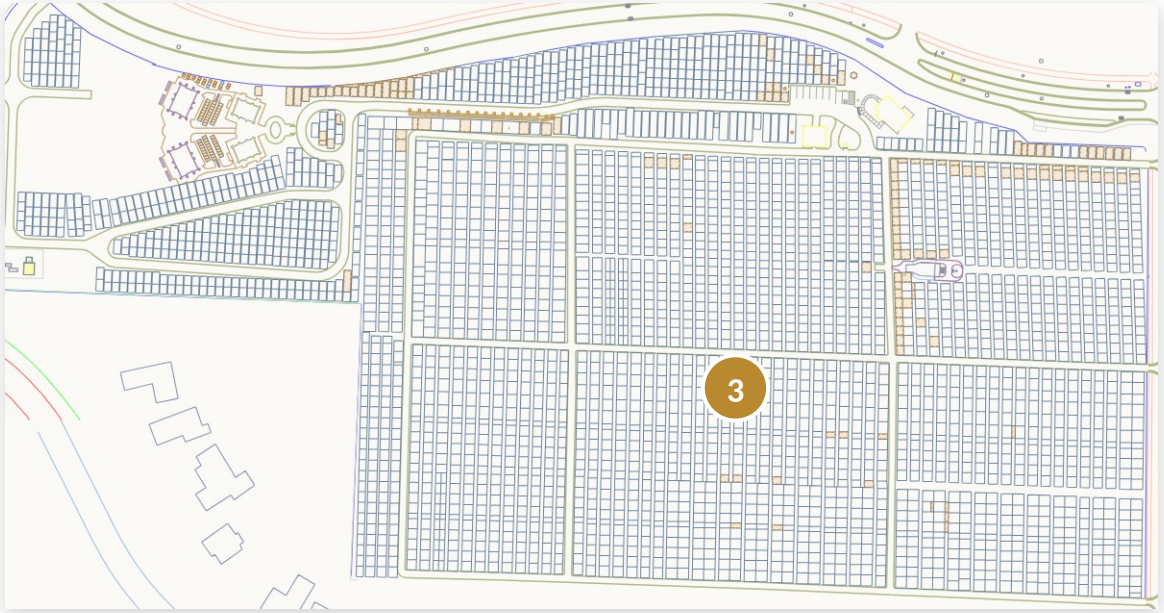
Annotations removed; roads and lot outlines kept

BEFORE: previous site



NOW: rebuilt map

Same map bounds in both screenshots



1. Dark "SECTION" title blocks sat over the lots.
2. Legend and descriptive notes were part of the picture.
3. Original roads and lot outlines retained, with stronger lines.

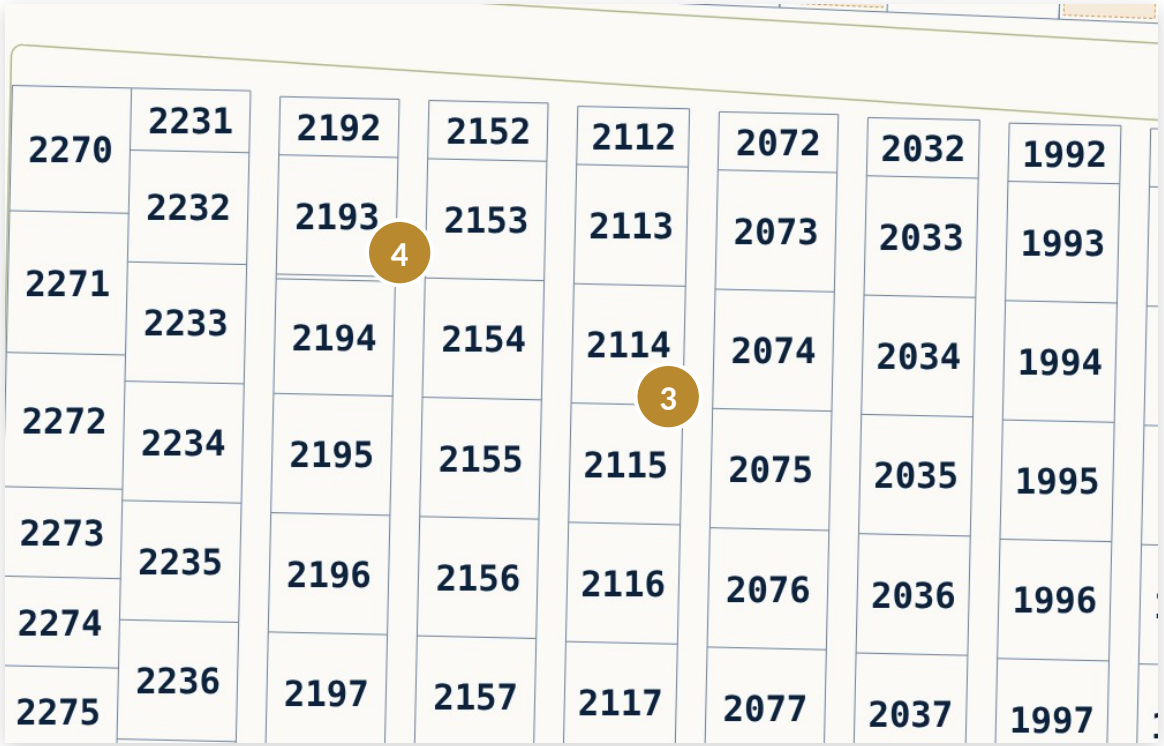
The first version drew new numbers over the old labels. In the rebuilt map the old text is never drawn. The original annotations can be switched on from Layers.

Confirmed H-numbers now fit inside their lot outlines

BEFORE Same lots, same zoom



NOW



1. Old numbers still showed beneath the new ones.
2. Heavy outlines pushed numbers against the lot ends.
3. One H-number per confirmed lot, inside its outline.
4. Legacy labels are not drawn in the rebuilt layer.

Labels follow the lot orientation, even on curves

BEFORE Same lots, same zoom



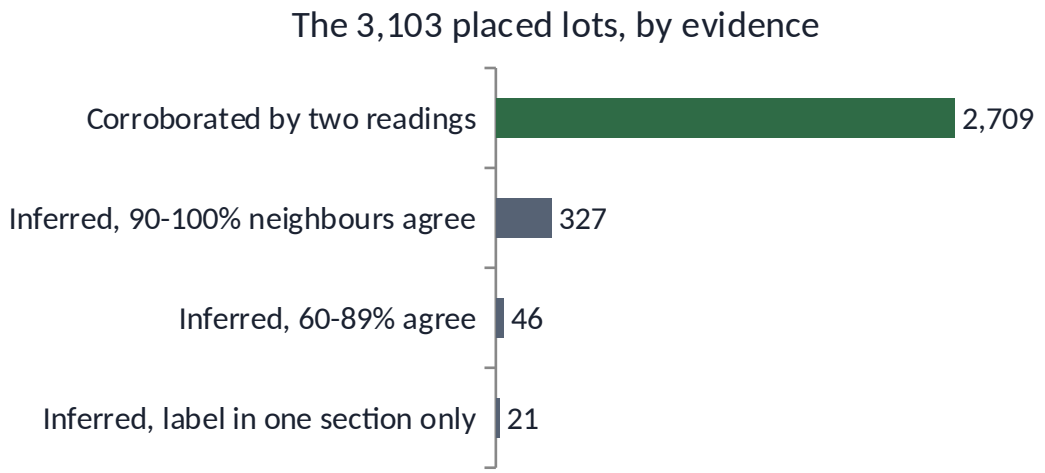
NOW

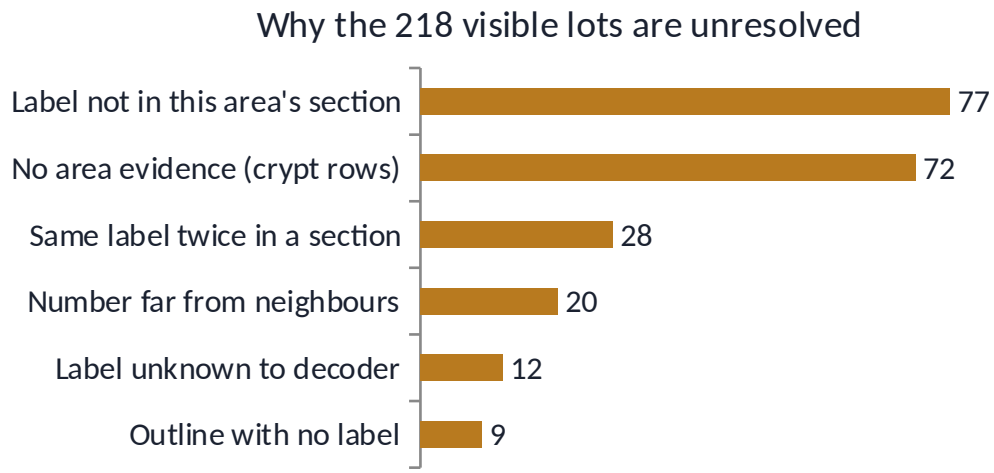


1. A "SECTION" block sits over lots 2894 to 2909.
2. Numbers cross lot edges.
3. Black smudges mark erased old labels.
4. Each number turns with its lot and stays inside it.
5. Narrow lots get smaller numbers: zoom in on screen. PDF detail pages print the smallest at 7.4 pt.

Where the lots stand: placed, unresolved, or unplaced

Status	Lots	What it means
Placed, section corroborated	2,709	A second, earlier reading of the drawing puts the label in the same legacy section.
Placed, section inferred	394	Section inferred from neighbouring lots. Not yet reviewed one by one.
Visible lot, number unresolved	218	An outline is on the drawing; the evidence does not support one H-number. Shaded, never numbered.
Decoder lot, no map position	68	Known to the decoder; no label resolves to it. 34 of these match a drawn lot counted in the 218.





The 218 and 68 overlap, so they should not be added together. No burial records and no verified grave counts are in the shared database yet.

Numbers stored as text changed the method

The key finding: the source drawing (an Illustrator .ai vector file) stores lot numbers as live text and lot outlines as separate lines. I rebuilt the pipeline around that, so numbers are read instead of recognized and nothing has to be covered up.

Earlier build: the drawing held numbers as strokes

Overall map and nine section pages (PDF) → Recognize each drawn character → Align the pages to the overall map → Decode to H-number

earlier readings kept as an independent cross-check of each label's legacy section

Current build: the drawing holds numbers as text

Read text and lines from the .ai file → Trace each lot outline → Match labels to lots; decode H-number → Check neighbours, duplicates, collisions → Render map, PDFs and site

Alignment: earlier pages were aligned to the overall map within 0.05 map units, and the earlier readings to the source drawing within about 0.1 (roughly 2% of a lot width). This transfers labels between drawings; it is not survey accuracy.

Tools, grouped by the job they did

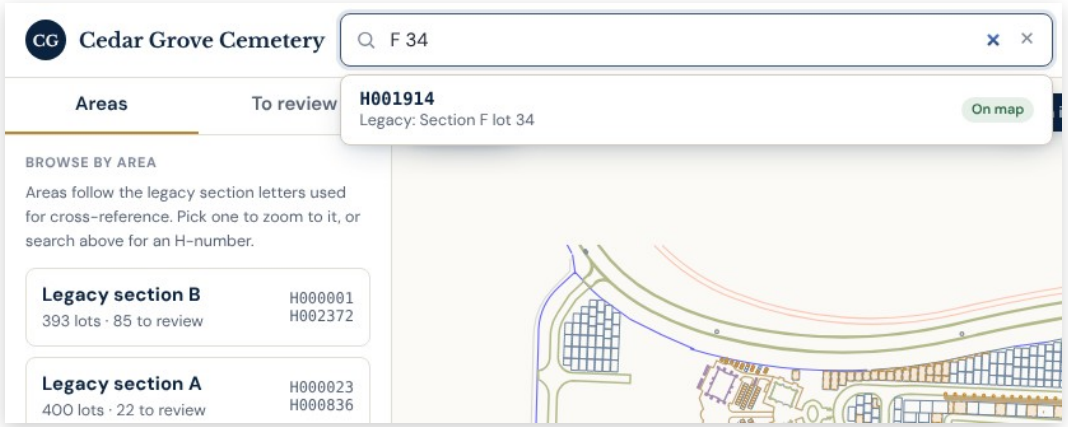
Job	Tools	Why it mattered
Read the source drawings	Python 3, PyMuPDF	Pull every line and every text label out of the .ai file, and write the PDFs.
Find and fit the lot shapes	OpenCV, NumPy, SciPy, Shapely	Trace each lot outline, match numbers to the right lot, and keep each number inside it.
Decode H-numbers	Project decoder (s4decoder1.py)	Turns "Section F, lot 34" into H001914. It is the only source of H-numbers.
Build the map and site	SVG, plain JavaScript, shared database	A sharp, zoomable map with lot panels. Records are stored so that only people with edit access can change them.
Check the result	Playwright, Chromium	Take the screenshots and run the automatic placement checks in a real browser.

Checks: no label-to-line collisions were detected in the tested screen and PDF renders. That supports readability; it does not prove a number belongs to the right physical lot.

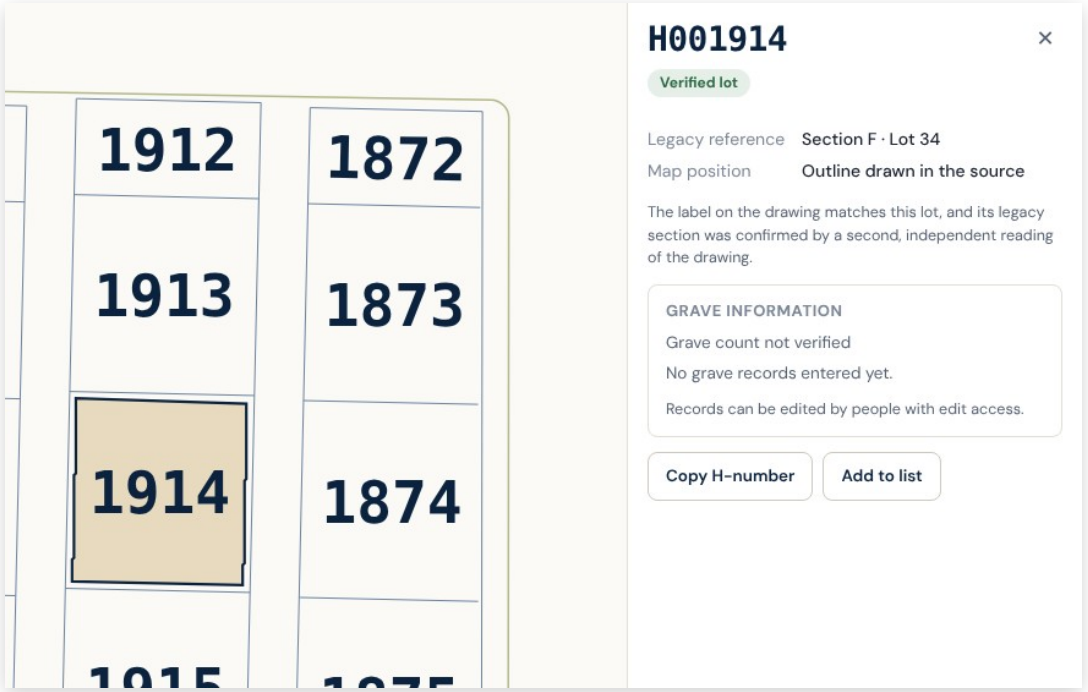
Formats: .ai (source) > JSON (working data) > SVG (map), PDF, CSV and Excel (.xlsx) (outputs).

One task from start to finish: find lot F 34 and open it

1 Type a legacy label or H-number



2 Select it: the lot is zoomed, highlighted, described



Working: tested in a browser

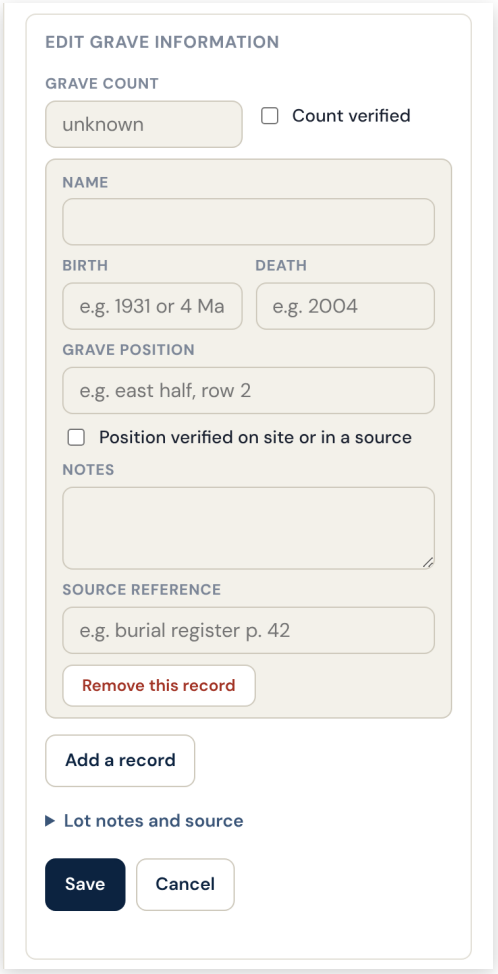
Search takes an H-number, a legacy lot, or a recorded name. Name search is implemented but cannot be shown until burial records exist.

Open decision: the sidebar still groups lots by legacy section letter, labelled as cross-reference. I kept it because no approved area grouping exists yet.

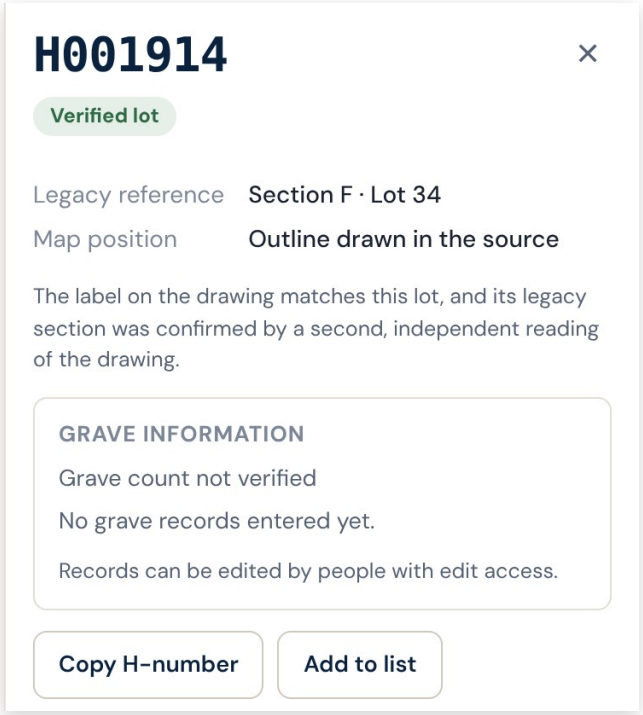
The panel stays beside the lot, so the selected lot is never covered. Back and Fit map return to earlier views.

Records, editing and exports: built, with live testing still to do

Editing: not tested live



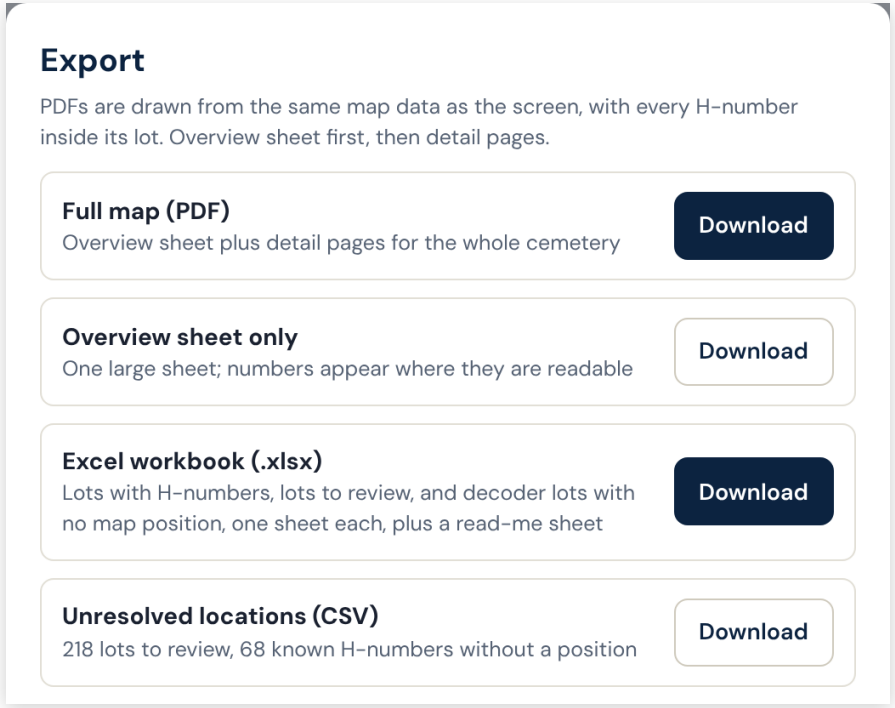
Viewing: working



Each record is one burial in the lot. Grave position is optional text, marked verified or not. Fields start empty and say so.

A standard viewer's write was refused in a service-level test. Editing is not yet tried with real editor accounts in the browser.

Files built; download untested



PDF, CSV and Excel (.xlsx) files are generated. Downloading them from the hosted page, and the record import, still need testing.

Next steps: who does what

My next development work

- Spot-check the 394 inferred-section lots, weakest 46 first
- Fix the exceptions I can: 9 unlabelled outlines, 3 lots too small for a number
- Validate the record import and report matched and skipped rows
- Test editing and downloads on the hosted page

Needed from cemetery staff

- A sample burial-record spreadsheet, even 20 rows
- Any lot-capacity or grave-position records
- Someone to rule on the numbering conflicts: B 373 to 380H, G lot 19, A 20H to 343H, B-side 395 to 414, crypt rows
- A short session to test editing with real accounts

To settle together

- Area names to show instead of legacy section letters
- Which resolved conflicts to accept
- Logo and where the site is hosted

Next review: remaining map exceptions, a sample record import, and verified PDF and Excel downloads.